Condensed Matter Physics I

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Exercise Sheet 1

Exercise 1 — *2D Bravais lattices* [easy]

(a) In two dimensions, there are five Bravais lattices. Name and sketch them.

(b) Give a unified description using the primitive vectors $\vec{a}_1$, $\vec{a}_2$ and the angle φ between them. Write the general lattice translation as

$$\vec{R} = n_1\,\vec{a}_1 + n_2\,\vec{a}_2, \qquad n_1, n_2 \in \mathbb{Z}.$$

(c) From the patterns in Fig. 1, decide which are Bravais lattices. For each Bravais lattice, name the type and sketch two different primitive cells (with basis vectors).

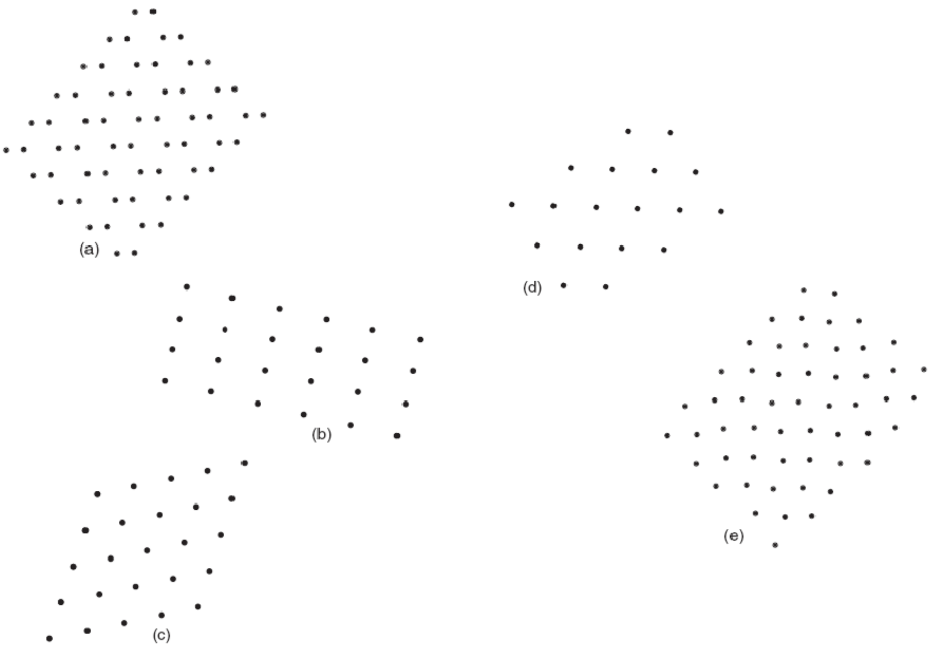

Figure 1: Find possible 2D Bravais lattices.

Exercise 2 — *Copper–oxygen planes* [moderate]

All copper oxide-based high T_C superconductors possess in their crystal structure as a central structural element copper-oxygen planes. The blue color atoms in the drawing (left in Fig. 2) are the Cu ions and the yellow color atoms are the O ions. The lattice constant is a in both directions (Cu ions form a square lattice). For the sake of simplicity we treat the problem as a two dimensional case.

(a) Define the rotation symmetry, Bravais lattice, a convenient set of primitive vectors, unit cell, and basis for the idealized CuO plane (all ions in plane).

(b) In La_2CuO_4, the O ions are displaced out of plane (alternating up/down). Discuss how this affects the symmetry, primitive cell and Bravais lattice. Can a still be used as the lattice constant?

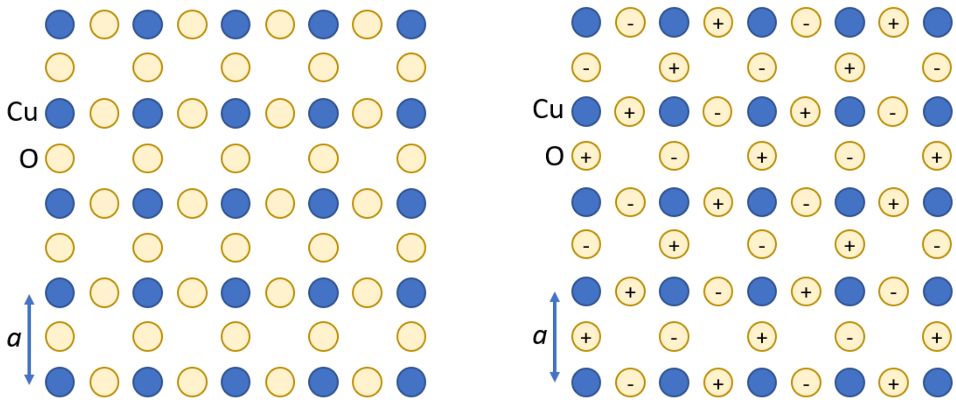

Figure 2: Left: CuO-planes of cuprate superconductors. Right: oxygen ions shifted up (+) or down (-) out of plane.

Exercise 3 — *Primitive vs. conventional cubic cell* [moderate]

A crystal has a basis of one atom per lattice point and primitive vectors (in 10^{-10} m):

$$\vec{a} = 3\,\hat{e}_x, \qquad \vec{b} = 3\,\hat{e}_y, \qquad \vec{c} = \tfrac{3}{2}\big(\hat{e}_x + \hat{e}_y + \hat{e}_z\big).$$

(a) Determine the Bravais lattice type.

(b) Calculate the volumes of the primitive cell and of the conventional cubic cell.

Exercise 4 — *Rotational symmetry* [moderate]

We investigate the admissible lattice symmetries for n-fold rotation axes, i.e. with a rotation angle $\varphi = 2\pi/n$. Which values of n are possible? To this end, consider four lattice points A, B, C, and D in the plane orthogonal to an n-fold rotation axis. Let A and B be nearest neighbours in a Bravais lattice at distance a, and assume that one of the n-fold rotation axes passes through them. A rotation about B maps A to C, and a rotation about A maps B to D. Show that the geometric constraints of the lattice allow only $n = 1, 2, 3, 4$, and 6.

Hint: Make a sketch. What conditions must the sides of the resulting trapezoid satisfy so that A, B, C, and D are lattice points?